

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium(Assignment 1)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 40 (each student assigned distinct questions) Total Marks: 100

ANNEXURE A

Assignment 1

Code of Conduct:

I Sameer Ahmed G (192421154) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sameer

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding				
Algorithm				
Code Implementation				
Competitive Performance				
Test Case Handling				

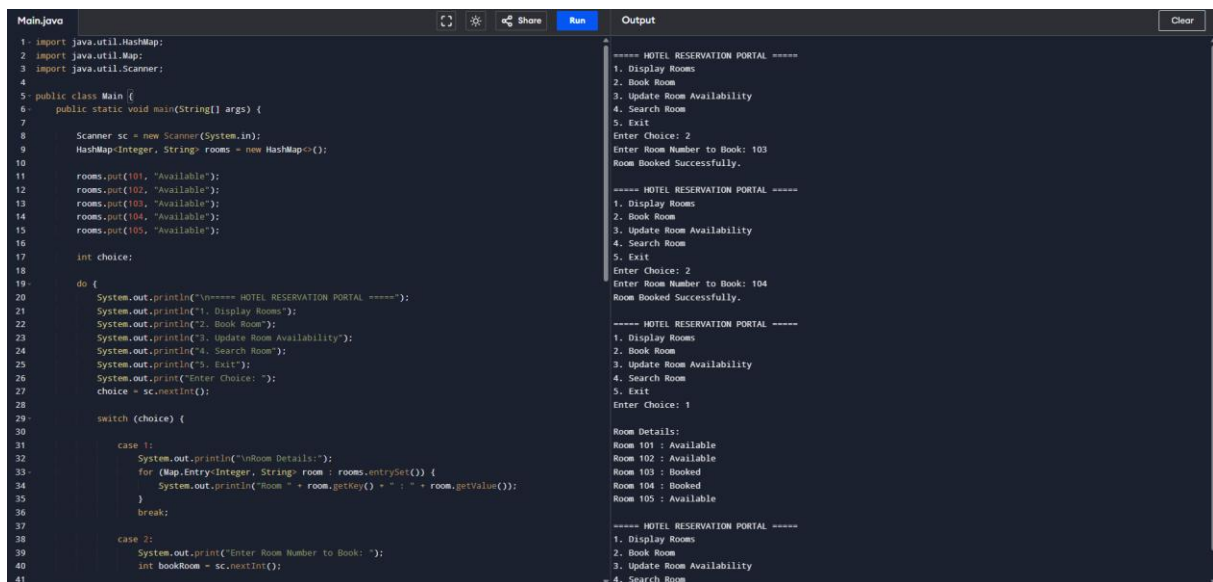
Question 22 – Hotel Reservation Portal

Problem Statement:

A hotel manages room bookings and customer details.

Questions

- Develop a Java program using **HashMap**.
- Update room availability.
- Compare HashMap and Hashtable.
- Explain reservation systems.



```
Main.java
1- import java.util.HashMap;
2- import java.util.Map;
3- import java.util.Scanner;
4-
5- public class Main {
6-     public static void main(String[] args) {
7-
8-         Scanner sc = new Scanner(System.in);
9-         HashMap<Integer, String> rooms = new HashMap<>();
10-
11-         rooms.put(101, "Available");
12-         rooms.put(102, "Available");
13-         rooms.put(103, "Available");
14-         rooms.put(104, "Available");
15-         rooms.put(105, "Available");
16-
17-         int choice;
18-
19-         do {
20-             System.out.println("\n===== HOTEL RESERVATION PORTAL =====");
21-             System.out.println("1. Display Rooms");
22-             System.out.println("2. Book Room");
23-             System.out.println("3. Update Room Availability");
24-             System.out.println("4. Search Room");
25-             System.out.println("5. Exit");
26-             System.out.print("Enter Choice: ");
27-             choice = sc.nextInt();
28-
29-             switch (choice) {
30-
31-                 case 1:
32-                     System.out.println("\nRoom Details:");
33-                     for (Map.Entry<Integer, String> room : rooms.entrySet()) {
34-                         System.out.println("Room " + room.getKey() + " : " + room.getValue());
35-                     }
36-                     break;
37-
38-                 case 2:
39-                     System.out.print("Enter Room Number to Book: ");
40-                     int bookRoom = sc.nextInt();
41-
42-                     if (rooms.containsKey(bookRoom) && rooms.get(bookRoom).equals("Available")) {
43-                         rooms.put(bookRoom, "Booked");
44-                         System.out.println("Room Booked Successfully.");
45-                     } else {
46-                         System.out.println("Room is not available or already booked.");
47-                     }
48-                     break;
49-
50-                 case 3:
51-                     System.out.print("Enter Room Number to Update: ");
52-                     int updateRoom = sc.nextInt();
53-
54-                     if (rooms.containsKey(updateRoom)) {
55-                         System.out.print("Enter new status (Available/Booked): ");
56-                         String newStatus = sc.next();
57-                         rooms.put(updateRoom, newStatus);
58-                         System.out.println("Room Updated Successfully.");
59-                     } else {
60-                         System.out.println("Room is not available for update.");
61-                     }
62-                     break;
63-
64-                 case 4:
65-                     System.out.print("Enter Room Number to Search: ");
66-                     int searchRoom = sc.nextInt();
67-
68-                     if (rooms.containsKey(searchRoom)) {
69-                         System.out.println("Room Status: " + rooms.get(searchRoom));
70-                     } else {
71-                         System.out.println("Room not found in the system.");
72-                     }
73-                     break;
74-
75-                 case 5:
76-                     System.out.println("Exiting the program. Goodbye!");
77-                     return;
78-
79-                 default:
80-                     System.out.println("Invalid choice. Please enter a valid option (1-5).");
81-                     break;
82-
83-             }
84-
85-             System.out.println("\n===== HOTEL RESERVATION PORTAL =====");
86-             System.out.println("1. Display Rooms");
87-             System.out.println("2. Book Room");
88-             System.out.println("3. Update Room Availability");
89-             System.out.println("4. Search Room");
90-             System.out.println("5. Exit");
91-             System.out.print("Enter Choice: ");
92-             choice = sc.nextInt();
93-
94-         } while (choice != 5);
95-     }
96- }
```

```
Output
===== HOTEL RESERVATION PORTAL =====
1. Display Rooms
2. Book Room
3. Update Room Availability
4. Search Room
5. Exit
Enter Choice: 2
Enter Room Number to Book: 103
Room Booked Successfully.

===== HOTEL RESERVATION PORTAL =====
1. Display Rooms
2. Book Room
3. Update Room Availability
4. Search Room
5. Exit
Enter Choice: 2
Enter Room Number to Book: 104
Room Booked Successfully.

===== HOTEL RESERVATION PORTAL =====
1. Display Rooms
2. Book Room
3. Update Room Availability
4. Search Room
5. Exit
Enter Choice: 1

Room Details:
Room 101 : Available
Room 102 : Available
Room 103 : Booked
Room 104 : Booked
Room 105 : Available

===== HOTEL RESERVATION PORTAL =====
1. Display Rooms
2. Book Room
3. Update Room Availability
4. Search Room
```

a),b) Code

```
import java.util.HashMap;
import java.util.Map;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        HashMap<Integer, String> rooms = new HashMap<>();

        rooms.put(101, "Available");
        rooms.put(102, "Available");
        rooms.put(103, "Available");
        rooms.put(104, "Available");
        rooms.put(105, "Available");

        int choice;

        do {
            System.out.println("\n===== HOTEL RESERVATION PORTAL =====");
            System.out.println("1. Display Rooms");
            System.out.println("2. Book Room");
            System.out.println("3. Update Room Availability");
            System.out.println("4. Search Room");
            System.out.println("5. Exit");
            System.out.print("Enter Choice: ");
            choice = sc.nextInt();

            switch (choice) {
```

case 1:

```
System.out.println("\nRoom Details:");  
for (Map.Entry<Integer, String> room : rooms.entrySet()) {  
    System.out.println("Room " + room.getKey() + " : " + room.getValue());  
}  
break;
```

case 2:

```
System.out.print("Enter Room Number to Book: ");  
int bookRoom = sc.nextInt();  
  
if (rooms.containsKey(bookRoom)) {  
    if (rooms.get(bookRoom).equals("Available")) {  
        rooms.put(bookRoom, "Booked");  
        System.out.println("Room Booked Successfully.");  
    } else {  
        System.out.println("Room Already Booked.");  
    }  
} else {  
    System.out.println("Room Not Found.");  
}  
break;
```

case 3:

```
System.out.print("Enter Room Number: ");  
int roomNo = sc.nextInt();  
sc.nextLine();  
  
if (rooms.containsKey(roomNo)) {  
    System.out.print("Enter Status (Available/Booked): ");  
    String status = sc.nextLine();  
    rooms.put(roomNo, status);  
}
```

```
        System.out.println("Room Status Updated.");
    } else {
        System.out.println("Room Not Found.");
    }
    break;

case 4:
    System.out.print("Enter Room Number: ");
    int search = sc.nextInt();

    if (rooms.containsKey(search)) {
        System.out.println("Room " + search + " : " + rooms.get(search));
    } else {
        System.out.println("Room Not Found.");
    }
    break;

case 5:
    System.out.println("Thank You!");
    break;

default:
    System.out.println("Invalid Choice.");
}

} while (choice != 5);

sc.close();
}
}
```

c) Compare HashMap and Hashtable

HashMap	Hashtable
Not synchronized	Synchronized
Faster	Slower
Allows one null key and multiple null values	Does not allow null keys or values
Introduced in Java Collections Framework	Legacy class
Not thread-safe	Thread-safe

d) Explain Reservation Systems

A reservation system is used to manage hotel room bookings efficiently. It stores customer information, room availability, booking details, and payment records. The system prevents double booking, updates room status automatically after reservation or cancellation, and enables quick search and management of customer reservations. It improves booking accuracy, reduces manual work, and enhances customer service.